

Programme	BSc Computer Science				
Course Title	Fundamentals of Data Science				
Type of Course	Vocational Minor				
Semester	II				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Basic understanding of computer science concepts. 2. Familiarity with data handling. 3. simple mathematical analysis.				
Course Summary	Data science is the domain of study that deals with vast volumes of data using modern tools and techniques to find unseen patterns, derive meaningful information, and make business decisions.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Identify the relevance and applications of computers in other disciplines with various data science applications.	R	C	Assignment / Instructor-created exams / Quiz
CO2	understanding of data science concepts and be capable of applying data science skills and interpret data science results	U	C	Assignment / Instructor-created exams / Quiz
CO3	Acquire logical thinking about evolution of data science	U	C	Assignment / Instructor-created exams / Quiz
CO4	How to use tools for acquiring, cleaning, analyzing, exploring, and visualizing data	Ap	P	Assignment / Instructor-created exams / Quiz
CO5	Learn to make data-driven inferences and decisions	Ap	P	Assignment / Instructor-created exams / Quiz
CO6	Able to perform data science processing, such as data import, data analysis, data visualization, and data modelling	Ap	P	Assignment / Instructor-created exams / Quiz

* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)
 # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P)
 Metacognitive Knowledge (M)

Detailed Syllabus

Module	Unit	Content	Hrs (45+30)	Mark
I		Introduction to Data Science	10	15
	1	Introduction to Data Science-Definition	2	
	2	Evolution of Data Science	2	
	3	Data Science Roles	3	
	4	Application of data sciences.	3	
II		Data Collection and Data Pre-Processing	10	15
	5	Data Collection Strategies	1	
	6	Data Pre-Processing Overview	2	
	7	Data Cleaning	2	
	8	Data Integration and Transformation	3	
	9	Data Reduction and Descrretization	2	
III		Data Analytics	12	20
	10	Descriptive Statistics	2	
	11	Mean, Standard Deviation	2	
	12	Skewness and Kurtosis	2	
	13	Box Plots	2	
	14	Pivot Table	2	
	15	Correlation Statistics	2	
IV		Data Model Devopment and Evaluation	13	20
	16	Simple and Multiple Regression	2	
	17	Model Evaluation using Visualization	2	
	18	Residual plot and distributional plot	2	
	19	Prediction and Decision Making	2	
	20	Model Evaluation techniques-	3	
	21	Supervised learning techniques	1	
	22	unsupervised learning techniques	1	
V		Practical: Introduction to data analysis tools in Python - Working with Pandas data frames - Basic plots using Matplotlib - Frequency distributions - Averages - Correlation and scatter plots - Correlation coefficient - Regression	30	

Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	3	-	1	-	2	-						
CO 2	3	-	1	-	1	-						
CO 3	3	-	2	-	1	-						
CO 4	2	-	2	-	2	-						
CO 5	1	-	2	-	2	-						
CO 6	1	-	2	1	2	-						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Practical Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓

CO 3		✓	✓	✓
CO 4		✓	✓	✓
CO 5		✓	✓	✓
CO 6	✓	✓		✓

Text books:

1. Jojo Moolayil, “Smarter Decisions : The Intersection of IoT and Data Science”, PACKT, 2016.
2. Cathy O’Neil and Rachel Schutt , “Doing Data Science”, O’Reilly, 2015.
3. David Dietrich, Barry Heller, Beibei Yang, “Data Science and Big data Analytics”, EMC 2013
4. Introduction to Data Science a Python approach to concepts, Techniques and Applications, Igual, L;Seghi’, S. Springer, ISBN:978-3-319-50016-4 2.
5. Data Analysis with Python A Modern Approach, David Taieb, Packt Publishing, ISBN-9781789950069